

CS425 Group Project Guideline

1. Objective

The project gives the teams opportunity to freely explore ideas and solutions in building a data-driven or non-data-driven conversational system. The specific goal and scope of the project shall be defined via open discussions among team members within each group. You can utilize and/or extend any methods and techniques covered in the class. You can work on a social chatbot or task-oriented bot or a mixture of two. The datasets and base source codes used shall be decided by your own.

You can choose any method, architecture and dataset to build your model for training and testing, **except using any LLM APIs**. It should meet the following executable conditions:

1. It can be trained and tested on a normally configured personal computer with or without GPU, **OR**
2. It can be trained and tested on Google Colab or on a GPU server in the SCIS.

You can either create a system from scratch or fine-tune the system based on existing packages, source codes or frameworks, but there shall be *new contributions* rather than just replicating an existing system by simply running open-source code with the same data. Essentially, we will look for your creative idea and how you apply the learned methods, techniques, tools to realize it.

2. Summary of Deliverables

The project component accounts for 35% (30% group: proposal/presentation/report; 5% individual: peer evaluation) of the final grade. Further breakdown is listed below.

Deliverable	Weight	Deadline	Description
Group formation	-	11 Sept 2026	Max. 5 members
Pre-proposal workshop	-	29 Sept 2026	Max. 15 min/group
Proposal	10%	02 Oct 2026	Max. 3 pages, min. font size 11
Final presentation	10%	10 Nov 2026	Max. 25 mins/group
Final report	10%	27 Nov 2026	Max. 15 pages, min. font size 11
Peer evaluation	5%	27 Nov 2026	Peer evaluation form will be provided

* Slightly relaxed as long as you can fix your group by the time of pre-proposal workshop

3. What next:

1. Form groups: **maximum 5 (or 6 in exceptional case subject to approval) members per group**. Please organize the groups among yourself before deadline. Talk with instructor and/or TA if you can't find a group.
2. In the pre-proposal work (**29 September 2026, Tuesday**), each group will spend around 15 mins to present and discuss your preliminary idea and plan. You can use slides or any form of discussion tool. This will **not be graded**, and is only for preparation of your proposal. Feedback will be provided during the workshop.
3. Submit a max 3-page proposal per group (including references, A4 paper, 11 pt font) explaining your idea as well as plan of method, data, and implementation. Make sure the name list and proposed duty of group members are also provided. (Due at **11:59pm, 02 October 2026, Friday**). (10%)

4. Each group will give a max. 25-min oral presentation showing your specific work in the project in the class of Week 13 (**10 November 2026, Tuesday**). (10%)
 - a. Overview: Introduce the motivation, scope and objectives of the project.
 - b. What did you do: Describe what your data and task(s) are in the teamwork.
 - c. How did you do it: Describe the method(s) you used to deal with your task(s), including data processing, model/architecture/framework, etc.
 - d. How good/bad is the outcome: What specific outcomes are observed, how did you make improvements.
 - e. Why did that happen: Analysis on pros/cons of the methods that result in such kind of outcomes.
 - f. Answer questions reasonably and clearly.

You should aim to have at least **90%** of your results completed by the time of final presentation. Your presentation will be graded based on your **content, clarity** and **question handling**. Feedback will be given to you on the spot to help you improve the final report.

5. Submit a final group report including the code and data in the Week 15 (Due at **11:59pm, 27 November 2026, Friday**). (10%) The report should be maximum **15 A4 pages** long (11 pt font) including all figures, references and appendices. The report should include:
 - a. The introduction of your project motivation, goals or hypothesis, together with any background necessary to understand your motivation, goal or hypothesis. What differences you made relative to the existing work your system is based upon (1%).
 - b. The description of your dataset(s), pre-processing, model and architecture, as well as implementation or fine-tuning details. (2%)
 - c. The settings of the key hyper-parameters if any. (2%)
 - d. The description of how to run your system for data pre-processing, training and testing. (2% -- the code should be runnable.)
 - e. Evaluation results with quantitative performance metrics, or qualitative analysis based on 15-20 input query sentences and their responses generated by your system. (2%)
 - f. The list of references. (1%)

Your final report will be graded based on its **interestingness, validity** and **soundness** and **clarity**.

6. Submit a peer evaluation form, in which each team member should assess the performance of every other team member based on the given template. No mark will be earned for those who do not provide assessment to others. Due at **11:59pm, 27 November 2026, Friday**. (5%)

All submissions must go to eLearn (a dropbox for submissions will be created).

Appendix

Some project ideas and resources (This is for your reference only. Other than these, you can freely explore and use any other resources online as you see feasible)

- Anyq: FAQ-based Question Answering System
<https://github.com/baidu/AnyQ>
- CakeChat: Emotional Generative Dialog System
<https://github.com/lukalabs/cakechat>

- Plato Research Dialogue System: This is the Plato Research Dialogue System, a flexible platform for developing conversational AI agents.
<https://github.com/uber-research/plato-research-dialogue-system>
- Conv Emotion: This repo contains implementation of different architectures for emotion recognition in conversations.
<https://github.com/declare-lab/conv-emotion>
- Rnnlg: RNNLG is an open source benchmark toolkit for Natural Language Generation (NLG) in spoken dialogue system application domains.
<https://github.com/shawnwun/RNNLG>
- Nndial: NNDial is an open source toolkit for building end-to-end trainable task-oriented dialogue models.
<https://github.com/shawnwun/NNDIAL>
- Mem2seq: Mem2Seq: Effectively Incorporating Knowledge Bases into End-to-End Task-Oriented Dialog Systems.
<https://github.com/HLTCHKUST/Mem2Seq>
- LIDA: Lightweight Interactive Dialogue Annotator.
<https://github.com/Wluper/lida>

A list of dialogue datasets for Chatbot building:

- CMU Question-Answer Dataset
<http://www.cs.cmu.edu/~ark/QA-data/>
- Ubuntu Dialogue Corpus
<https://www.kaggle.com/rtatman/ubuntu-dialogue-corpus>
- Customer Support on Twitter:
<https://www.kaggle.com/thoughtvector/customer-support-on-twitter>
- Cornell Movie Dialogue Corpus:
https://www.cs.cornell.edu/~cristian/Cornell_Movie-Dialogs_Corpus.html
- The Conversational Intelligence Challenge 2 (ConvAI2) Dataset:
<http://convai.io/2017/data/>
- And many others to be discovered by yourself ...